Online PES Decoder Learning in Neuromorphic Residual Control of UAV Quaternion Tracking

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***Abstract*— Quadrotor attitude controllers must track time-varying references while respecting coupled rotor-force limits and coping with modeling, measurement uncertainty, and disturbances. We present an online neuromorphic residual architecture based on six continuous coupled conductance-based membrane-gate units that generate temporal features, and eighteen decoder weights adapt online through prescribed error sensitivity (PES). The decoded correction is bounded and added to a saturated PD quaternion-tracking controller before the rotor-force allocation, preserving classical feedback. A conditional local input-to-state analysis gives explicit bounds on the attitude perturbation and residual authority under stated inertia and allocation assumptions. Across ten matched cases on two airframes, the residual lowers mean tracking error by 17.3 to 60.6 percent against that fixed baseline, and by a further 58 percent in geometric mean against a feedforward PD at the same gains, every case improving and the airframes agreeing to 0.2 points. Against a bandwidth-tuned feedforward PD the benefit narrows to 36 and 7 percent and reverses on the abrupt-step mission, where the baseline already saturates the rotors and the lagged correction subtracts authority. These simulation results identify persistent temporal activity combined with online decoder learning as the principal mechanism behind the observed improvement.**

I. Introduction

Quadrotor attitude controllers must track time-varying references while respecting coupled rotor-force limits under modeling uncertainty, measurement noise, and disturbances. Classical feedback provides an interpretable stabilizing backbone, but unmodeled effects and abrupt disturbances can leave structured tracking errors that are difficult to remove without retuning or increasing actuator demand. Residual control offers a complementary approach: a learned correction adapts online while preserving the classical feedback law and actuator interface.

Residual learning has been combined with classical controllers in robotics [1], including geometric $\mathcal{L}_1$ quadrotor augmentation with performance guarantees [2] and proportional–integral–derivative (PID) attitude control augmented by a proximal policy optimization (PPO) residual [3]. Neuromorphic control provides another route for stateful adaptive corrections, and brain-inspired robust UAS designs pair model-based feedback with learned terms, from limbic-system controllers [4], [5] to neural weight reset under impulsive disturbances [6] and perturbation estimation on small quadrotors [7]. Nengo supports the Neural Engineering Framework [8] and online decoder learning [9], most often through the prescribed error sensitivity (PES) rule [10], a local error-driven update [11]. PES-based adaptive control has been demonstrated on manipulators [12], [13] and in self-tuning spiking PID controllers for free-flying UAS trajectory tracking [14], [15]. However, continuous membrane–gate features for actuator-aware quaternion residual control, and the separate roles of neuron dynamics and decoder learning, remain insufficiently characterized.

We study an online neuromorphic residual controller for quaternion attitude tracking with individual rotor-force saturation. A quaternion proportional–derivative (PD) controller supplies the baseline torque, six continuous coupled membrane–gate (CMG) units generate temporal features, and an eighteen-parameter decoder adapts online through a PES-style update. The bounded decoder output is added to the saturated baseline before common rotor-force allocation; setting the residual authority to zero therefore recovers the baseline exactly. We compare the residual with both the original fixed PD controller and a tuned PD controller with desired-rate feedforward, separating improvement over a basic reference from additional benefit beyond a stronger classical baseline.

The paper's contributions are:

1) a fully specified CMG–PES residual architecture with paired-current encoding, continuous state dynamics, online decoder adaptation, bounded readout, and explicit rotor-force allocation, Fig. 1;
2) analytic bounds on internal states, decoder weights, and command deviation, including exact baseline recovery at zero residual authority;
3) a controlled ablation separating conductance modulation, decoder learning, and residual authority, together with a matched neuron-family comparison; and
4) a two-platform, five-mission simulation study with measurement-noise and repeated-disturbance experiments that evaluates the online residual as the classical baseline is strengthened.

Each CMG unit couples a leaky membrane state to a gating state that modulates its effective conductance, combining LIF-like integration with Hodgkin–Huxley-inspired gating dynamics [16]. The controller uses the resulting continuous, non-spiking trajectory as a temporal feature basis; it does not require spike events or claim neuromorphic-hardware energy savings. In the tested regime, online decoder adaptation is the primary source of improvement, whereas conductance modulation has a comparatively small effect. Over a feedforward PD at the same gains the residual helps on both airframes alike; once that baseline is also bandwidth-tuned,

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the benefit narrows and becomes platform-dependent, and it reverses where the baseline saturates its rotors.

The remainder of the paper is organized as follows. Section II defines the attitude-tracking problem and actuator interface. Section III presents the CMG–PES architecture and learning rule. Section III establishes the boundedness properties. Section IV reports the baseline comparisons, ablations, neuron-family benchmark, noise experiments, and repeated-disturbance results. Section V summarizes the findings and limitations. Code, model files, and run archives will be available at `https://github.com/your-repo`.

II. Attitude Baseline and Actuator Interface

The objective is to reduce shortest-path attitude error for a prescribed $q_d(t)$ while respecting rotor-force limits. Baseline and augmented controllers share the state measurements, height loop, and allocator; Section III modifies only the requested attitude torque.

A. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ be the scalar-first Hamilton quaternion mapping body to inertial coordinates, and let $\boldsymbol{\omega}$ be the body angular velocity. The rotational dynamics are [17]

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

where $\boldsymbol{J} \succ 0$ is the body-frame inertia, $\boldsymbol{u}$ is the commanded body torque, and $\boldsymbol{\tau}_{\mathrm{act}}$ is the torque realized after rotor allocation and force clipping; hence $\boldsymbol{\tau}_{\mathrm{act}} = \boldsymbol{u}$ when the request is feasible and generally $\boldsymbol{\tau}_{\mathrm{act}} \neq \boldsymbol{u}$ otherwise. The terms $\boldsymbol{\tau}_{\mathrm{p}}$ and $\boldsymbol{\tau}_{\mathrm{d}}$ denote modeled passive and externally applied disturbance torques, respectively.

Using the corresponding passive quaternion $q = Q^{-1}$ [18], define

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T} \otimes q. \tag{3}$$

The desired attitude q_d uses the same convention. Define the attitude error

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{4}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\,\mathrm{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The metric θ_q is invariant to the quaternion sign. The sign selection is nonsmooth at $q_{e,0} = 0$, and no continuous feedback globally stabilizes rigid-body attitude [19], [20]: global tracking needs hybrid [21] or geometric-algebra [22], [23] formulations, so the analysis below is local.

For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^\mathsf{T}]^\mathsf{T}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^\mathsf{T}\boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2}\boldsymbol{q}_{e,v}^\mathsf{T}\boldsymbol{\omega}_e. \tag{8}$$

These are kinematic identities (no convergence implied). Reference Euler triples parameterize q_d in this passive convention.

B. Physical Inertia and Implemented PD Gains

Let $\boldsymbol{J} \succ 0$ denote the rigid-body inertia about the center of mass, expressed in body axes [17]. When principal moments $\boldsymbol{j}_{\mathrm{p}}$ are specified in a rotated principal-axis frame, the physical body-frame tensor is

$$\boldsymbol{J} = R_{BI}\,\mathrm{diag}(\boldsymbol{j}_{\mathrm{p}})R_{BI}^\mathsf{T}, \tag{9}$$

for the rotation from principal axes to body axes R_{BI}. The implementation sets axiswise gains using the principal moments in their reported ordering, $\boldsymbol{j}_{\mathrm{tune}}$, not necessarily $\mathrm{diag}(\boldsymbol{J})$ (Table I).

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n, \tag{10}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{11}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^\mathsf{T}$ rad/s, $\zeta = 0.95$, and $\mathrm{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. The requested-torque box is $\mathcal{U} = \prod_{i=1}^3 [-\ell_i, \ell_i]$; $\mathrm{clip}_{[a,b]}$ also acts componentwise, and $\oslash$ denotes componentwise division. We impose componentwise torque limits $\boldsymbol{\ell} \in \mathbb{R}^3_{>0}$ through $\mathrm{sat}_{\boldsymbol{\ell}}(\cdot)$. In our implementation, $\boldsymbol{\ell}$ is chosen proportional to the diagonal gain-scheduling vector $\boldsymbol{j}_{\mathrm{tune}}$ via $\ell_i = \max(j_{\mathrm{tune},i}\nu_i, 10^{-6}\ \mathrm{N\,m})$, with acceleration-scale vector $\boldsymbol{\nu} = (12, 12, 10)^\mathsf{T}$ rad/s^2. These controller-side bounds are not the rotor-feasible torque set. The law is quaternion feedback [24]; $\boldsymbol{\omega}_n$ is a tuning parameter rather than a claimed set of coupled closed-loop natural frequencies. Equation (11) damps the absolute body rate, whereas (8) defines the rate error $\boldsymbol{\omega}_e$. A residual that corrects a baseline missing that term would be credited with work the baseline could do itself, so the study also carries the feedforward law

$$\boldsymbol{u}_0^{\mathrm{ff}} = \mathrm{sat}_{\boldsymbol{\ell}}\big(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot (\boldsymbol{\omega} - \boldsymbol{\omega}_d)\big), \tag{12}$$

with $\boldsymbol{\omega}_d$ obtained by central differences of the reference, and a bandwidth-scaled variant $\boldsymbol{\omega}_n \mapsto \rho\boldsymbol{\omega}_n$. Gyroscopic cancellation remains omitted. The resulting gains are vehicle-specific through $\boldsymbol{j}_{\mathrm{tune}}$, even though $\boldsymbol{\omega}_n$ and ζ are shared; the same fixed gains are then retained for PD and its CMG augmentation. The local result in Section III-D uses these anisotropic gains; it does not replace them with a scalar-gain controller.

C. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{13}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^\mathsf{T}$ and the actuator's local reaction torque $\kappa_i f_i$ [25]. Here B maps rotor thrusts to total thrust and body torque; κ_i has units of meters, not angular velocity. Given a commanded collective thrust T and body-torque command $\boldsymbol{u}$, the corresponding inverse-allocation and

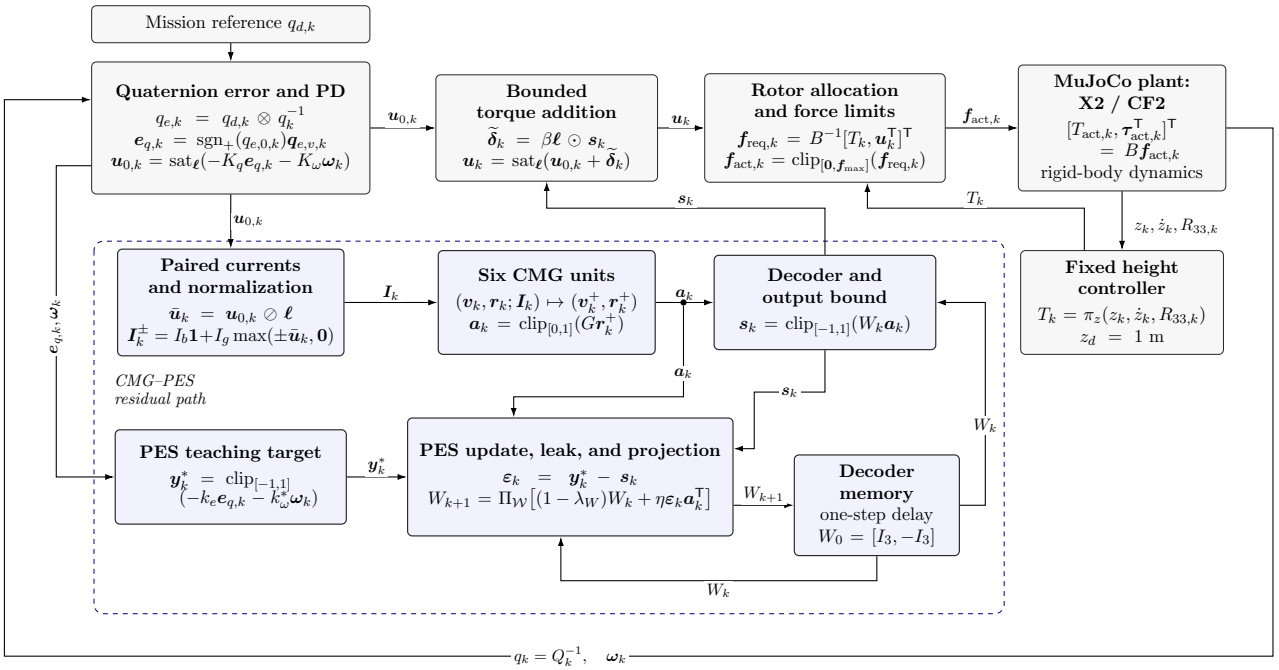


Fig. 1. Online PES decoder learning CMG-PES Neuromorphic Residual Control architecture



Fig. 2. Vehicles and simulation models. Left: CF2 hardware and MuJoCo model. Centre: X2 hardware and MuJoCo model. Right: both geometries at relative scale. The axis triads are body-frame axes: they are rigidly attached to the vehicle and therefore rotate with it relative to the inertial (world) frame (cf. Q: body $\to$ inertial in Sec. II).

clipping model is

$$\begin{aligned} f_{\mathrm{req}} &= B^{-1}[T, u^\mathsf{T}]^\mathsf{T}, \\ f_{\mathrm{act}} &= \mathrm{clip}_{[0, f_{\max}]}(f_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \tau_{\mathrm{act}}^\mathsf{T}]^\mathsf{T} &= B f_{\mathrm{act}}. \end{aligned} \tag{14}$$

Here $u = u_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{-1}[T, u^\mathsf{T}]^\mathsf{T} \in [0, f_{\max}]$; for four rotors B is square and invertible, so $B^{-1} = B^\dagger$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [26].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0, 2.5Mg]} \left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)} \right). \tag{15}$$

Here g is the gravitational acceleration. The height loop reads altitude and vertical velocity; the denominator compensates tilt. The guard 0.35 prevents near-singular tilt compensation as $R(Q)_{33} \to 0$ (i.e., near 90° tilt) and corresponds to limiting the effective compensation to about $\arccos(0.35) \approx 69.5^\circ$. Its upper bound does not supersede individual rotor limits. The CMG path has no direct collective output, but can affect height through attitude, tilt compensation, and allocation saturation. The resulting shared plant interface isolates changes in requested attitude torque while preserving the same physical simulation for both controllers.

III. CMG State Dynamics and Online Decoder Adaptation

This section specifies the residual architecture in state-space form and the online decoder update, then states the boundedness properties that follow from the resulting saturations and projections.

A. Encoding, States, and Numerical Update

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{u} &= u_0 \oslash \ell, \\ I^\pm &= I_b \mathbf{1} + I_g \max(\pm \bar{u}, \mathbf{0}), \\ I &= [(I^+)^\mathsf{T}, (I^-)^\mathsf{T}]^\mathsf{T}. \end{aligned} \tag{16}$$

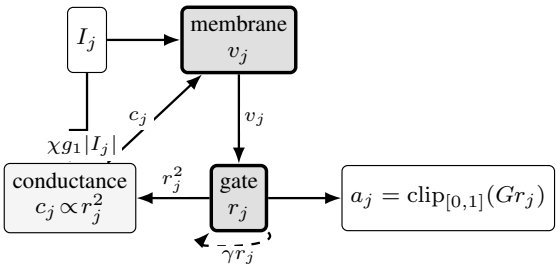


Fig. 3. Coupling topology of one CMG unit, cf. (19). The membrane sets the gate target; the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$ ($\chi = 1$ makes that shunt input dependent). The dynamics are continuous: no spike, no reset, and only a_j leaves the unit.

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{17}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{18}$$

$$c_j = (g_0 + \chi g_1|I_j|)r_j^2, \qquad \chi \in \{0, 1\}. \tag{19}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 3). Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{20}$$

$$r_{j,k}^{+} = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{21}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1|I_{j,k}|)(r_{j,k}^{+})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k}v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v_{j,k}^{+} = e^{-\lambda_{j,k}h}v_{j,k} + (1 - e^{-\lambda_{j,k}h})d_{j,k}/\lambda_{j,k}. \tag{22}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v_{j,k}^{+}, r_{j,k}^{+})$. The exponential-Euler step $r_{j,k}^{+} = e^{-h/\tau_r}r_{j,k} + (1 - e^{-h/\tau_r})b_{j,k}$ is convex for every $h > 0$ without a clip [27], and changes only the weight on $r_{j,k}$ (0.905 vs. 0.9 at $h/\tau_r = 0.1$); all results use (21).

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G\boldsymbol{r}_k^{+}), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k\boldsymbol{a}_k). \end{aligned} \tag{23}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. With $G = 10^5$ the activity saturates once $r_j \geq 10^{-5}$, so G is a primary design hyperparameter for how much of r_j's dynamic range reaches the decoder. The teaching target and local update are

$$\boldsymbol{y}_k^{*} = \mathrm{clip}_{[-1,1]}(-k_e\boldsymbol{e}_{q,k} - k_\omega^{*}\boldsymbol{\omega}_k), \tag{24}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}_k^{*} - \boldsymbol{s}_k, \tag{25}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta\boldsymbol{\varepsilon}_k\boldsymbol{a}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{26}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m\times 2m}$. The coefficients η and λ_W are per-step (so changing h changes the effective learning dynamics). PES is a local error-driven update toward the prescribed target $\boldsymbol{y}_k^{*}$; it does not differentiate through the plant or claim to minimize a closed-loop cost.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta\boldsymbol{\ell}\odot\boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \quad 0 \leq \beta \leq 1. \tag{27}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

We first state the boundedness properties that follow directly from the architecture: internal-state bounds, a uniform command-deviation bound set by residual authority, and projection-based bounds on decoder weights.

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table I, $r_{j,0} \in [0, 1]$, and $0 < h \leq \tau_r$, the gate remains in $[0, 1]$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. The unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0, 1]$. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \geq 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}. Equation (22) is a convex combination of $v_{j,k}$ and d_j/λ_j. Induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \leq \beta\ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness [28] and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \leq |\beta\ell_i s_{k,i}| \leq \beta\ell_i$. The recovery statements follow by substitution. □

The same allocator and state then give the same actuator output when $\beta = 0$. The proposition does not establish state safety, rotor headroom, stability, or a bound on tracking improvement. A bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (26) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta\|\boldsymbol{a}\|_2^2)\boldsymbol{\varepsilon}_k. \tag{28}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta\|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1}-W_k)\boldsymbol{a} = \eta\varepsilon_k\|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (28). The scalar multiplier has modulus less than one under the stated condition. □

This is the constant-feature PES result [11], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \le 2m$, $0 < \eta < 1/m$ is sufficient for that restricted model with nonzero activity; changing targets, loss of excitation, clipping, and leakage invalidate a direct application to the experiment. No unique identification of all decoder weights follows from one fixed regressor.

D. Allocation-Aware Local Stability

Let $E = [\mathbf{0}_{3\times1}, I_3]^\mathsf{T}$, $H = E^\mathsf{T}$, and let $\mathcal{A}_T(\boldsymbol{u}) = HB\,\mathrm{clip}_{[0,\boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T})$ be the realized torque at collective T. For two commands at the *same state and collective*, scalar clipping has secant slopes $d_i \in [0,1]$. Consequently,

$$\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0) = HBDB^{-1}E\boldsymbol{\delta}, \tag{29}$$

$$\|\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0)\|_2 \le L_A\beta\|\boldsymbol{\ell}\|_2, \tag{30}$$

where $D = \mathrm{diag}(d_i)$ and $L_A = \max_{D\in\mathcal{D}}\|HBDB^{-1}E\|_2$. $\mathcal{D}$ is the set of sixteen diagonal matrices with entries in $\{0,1\}$, and the vertex maximum bounds all intermediate D by convexity. Numerically $L_A = 4.53$ (X2) and 3.33 (CF2), inflated because small $|\kappa_i|$ turn clipped yaw requests into roll/pitch torque; if the whole box is rotor-feasible at the operating collective (CF2 at hover, not X2), $D = I$ and 1 replaces L_A. A yaw-weighted norm $\Lambda = \mathrm{diag}(1,1,c_\psi)$ shrinks the constant only if Theorem 1 is restated in those coordinates. This holds even with rotor clipping, but is not a bound between two different closed-loop trajectories. It also does not bound the height error.

For a constant reference and a smooth shortest-path branch, let $\boldsymbol{x} = [\boldsymbol{e}_q^\mathsf{T}, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$ and $K_q = \mathrm{diag}(\boldsymbol{k}_q)$, $K_\omega = \mathrm{diag}(\boldsymbol{k}_\omega)$. Write the continuous-time attitude subsystem as

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x}) + F\boldsymbol{d},$$
$$A = \begin{bmatrix} 0 & \frac{1}{2}I_3 \\ -\boldsymbol{J}^{-1}K_q & -\boldsymbol{J}^{-1}K_\omega \end{bmatrix}, \qquad F = \begin{bmatrix} 0 \\ \boldsymbol{J}^{-1} \end{bmatrix}, \tag{31}$$

where $\boldsymbol{d} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} + K_q\boldsymbol{e}_q + K_\omega\boldsymbol{\omega}$ is the total deviation from ideal PD torque. It includes allocation and command-saturation errors, not just the neural request. The nonlinear remainder is

$$\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} \frac{1}{2}[(\sqrt{1-\|\boldsymbol{e}_q\|^2}-1)I_3 + [\boldsymbol{e}_q]_\times]\boldsymbol{\omega} \\ -\boldsymbol{J}^{-1}[\boldsymbol{\omega}]_\times\boldsymbol{J}\boldsymbol{\omega} \end{bmatrix} = O(\|\boldsymbol{x}\|^2), \tag{32}$$

with $[\boldsymbol{z}]_\times\boldsymbol{y} = \boldsymbol{z}\times\boldsymbol{y}$; norms use fixed coordinates with rates in rad/s. Since $\boldsymbol{\omega}\times\boldsymbol{J}\boldsymbol{\omega} = \boldsymbol{\omega}\times(\boldsymbol{J}-\mu I_3)\boldsymbol{\omega}$ with $\mu = \frac{1}{2}(\lambda_{\max}+\lambda_{\min})(\boldsymbol{J})$, the quadratic-growth constant is $L_r = \frac{\sqrt{2}}{4} + \frac{1}{2}(\kappa(\boldsymbol{J})-1)$, i.e., 1.048 (X2) and 0.529 (CF2), above sampled suprema 0.483 and 0.283.

Theorem 1 (Conditional local practical stability)**.** *Let* $\boldsymbol{J}, K_q, K_\omega \succ 0$*, let* q_d *be constant, and use the smooth branch in* (31)*. Choose* $Q_L \succ 0$ *and the solution* $P \succ 0$ *of* $A^\mathsf{T}P + PA = -Q_L$*. On* $\|\boldsymbol{x}\| \le r < 1$*, suppose* $\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2$ *and choose* r *so that* $c = \lambda_{\min}(Q_L) - 2\|P\|L_r r > 0$*. Set* $p_- = \lambda_{\min}(P)$*,* $p_+ = \lambda_{\max}(P)$*,* $b = \|PF\|$*, and* $V = \boldsymbol{x}^\mathsf{T}P\boldsymbol{x}$*. If*

$$V(0) \le p_- r^2, \qquad \|\boldsymbol{d}(t)\| \le \bar{d} < \frac{cr}{2b}\sqrt{\frac{p_-}{p_+}}, \tag{33}$$

then $V \le p_- r^2$ *is forward invariant and*

$$V(t) \le e^{-\alpha t}V(0) + \frac{4p_+b^2}{c^2}\bar{d}^2, \qquad \alpha = \frac{c}{2p_+}. \tag{34}$$

Thus attitude regulation is locally input-to-state stable; zero perturbation gives local exponential convergence of the plant state.

Proof. The linear system has energy $\frac{1}{2}\boldsymbol{\omega}^\mathsf{T}\boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{e}_q$ with derivative $-\boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega}$. Its only invariant zero-derivative trajectory is the origin, so by LaSalle's invariance principle [29] A is Hurwitz and $P \succ 0$ exists; K_q is symmetric, so products of inertia do not enter. Equations (31)–(32) give

$$\dot{V} \le -c\|\boldsymbol{x}\|^2 + 2b\|\boldsymbol{x}\|\bar{d} \le -\frac{c}{2}\|\boldsymbol{x}\|^2 + \frac{2b^2}{c}\bar{d}^2.$$

On $V = p_-r^2$, $\|\boldsymbol{x}\| \ge r\sqrt{p_-/p_+}$, so (33) makes the first bound strictly negative. The ellipsoid stays inside the radius-r ball; integration of $\dot{V} \le -\alpha V + 2b^2\bar{d}^2/c$ gives (34). □

If the same-state baseline deviation from ideal PD is bounded by $\bar{d}_0$, then (30) permits $\bar{d} = \bar{d}_0 + L_A\beta\|\boldsymbol{\ell}\|$ in the theorem, *provided* (33) holds. This is a sufficient local robustness condition, independent of the particular PES trajectory, not a PES convergence proof. Maximizing the admissible bound over r gives $r^* = \min\{\lambda_{\min}(Q_L)/(4\|P\|L_r), 1\}$ and, when $r^* < 1$, $\bar{d}_{\max} = \lambda_{\min}^2(Q_L)\sqrt{p_-/p_+}/(16\,b\,\|P\|\,L_r)$ with $\beta < \beta_{\max} = (\bar{d}_{\max} - \bar{d}_0)/(L_A\|\boldsymbol{\ell}\|_2)$. With $Q_L = I$ and $\bar{d}_0 = 0$ at hover this gives $r^* = 5.31\times10^{-2}$ (X2), 1.05×10^{-1} (CF2), and $\beta_{\max} = 9.69\times10^{-5}$ (X2) and 6.43×10^{-4} (CF2; 2.14×10^{-3} with exact allocation). The experiments use $\beta = 0.5$, far outside this conservative certificate: stability of the reported runs is observed, not certified. Moving references, the 180° sign boundary, sampled-data/solver error, and full translational stability remain outside this result; sample-and-hold error would need its own bound within $\boldsymbol{d}$.

a) Nengo realization.: In Nengo 4.1.0 a custom `NeuronType` exposes membrane and gate states, and direct current injection bypasses NEF gain/intercept fitting; unfiltered CMG activity uses $\eta_N = \eta(2m)/h$ and error $\boldsymbol{s} - \boldsymbol{y}^*$, with leak on pre-step weights. Its teaching graph adds a one-step latency, so the weight sequences differ; over ten direct/Nengo cases the maximum relative mean-error difference is 2.013×10^{-6}: tested numerical agreement, not formal equivalence.

IV. Experimental Protocol and Cross-Mission Results

A. Models, Parameters, and Measurements

MuJoCo provides rigid-body simulation with site-applied actuator wrenches [30]. Two Menagerie-derived multirotor

TABLE I

MODEL/INTERFACE PARAMETERS WITH MANUFACTURER CONTEXT (TOP) AND THE CANONICAL 3-CHANNEL CMG–PES PARAMETERS (BOTTOM). NEURAL STATES AND CURRENTS ARE NORMALIZED MODEL VARIABLES; LEARNING COEFFICIENTS ARE PER STEP. MASS AGREES WITH PUBLISHED CF2/X2 SPECIFICATIONS [33], [34]; FORCE LIMITS, INERTIAS, AND REACTION COEFFICIENTS ARE SIMULATION PARAMETERS, NOT IDENTIFIED HARDWARE RATINGS.

Parameter	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction magnitude $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Rotor force limit (N)	$.0875 \to .13125$	13
Total force limit (N)	$.350 \to .525$	52
Main-test thrust/weight	1.982	4.001
$\boldsymbol{j}_{\text{tune}}$ (10^{-5} kg m^2)	$(2.3951, 2.3951, 3.2347)$	—
$\boldsymbol{j}_{\text{tune}}$ (10^{-2} kg m^2)	—	$(6.0711, 3.6468, 2.5412)$
Rigid-body integrator	RK4	Euler
Experiment timestep (s)	0.0005	0.0005
Manufacturer thrust data	.568 N[a]	Not published[b]

[a]Historical CF2 measurement at 93.75% PWM using an external power supply [35]; not a continuous battery-powered rating. The modeled .525 N is below that observation. [b]Not reported in the reviewed X2 datasheet. Neither model includes motor lag; a historical CF2 motor step took about 180 ms [36], illustrating an omitted hardware effect.

CMG–PES	Value		Value
h, duration	0.0005 s, 45 s	$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
τ, τ_r	0.05 s, 0.005 s	I_b, I_g, G	$0.75, 1.25, 10^5$
$v_{\text{rest}}, v_{\text{reset}}, v_{\text{th}}$	$0, 0, 1$	β, η	$0.5, 8 \times 10^{-4}$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$	$\lambda_W, W_{\max}, k_e, k_\omega^*$	$10^{-6}, 6, 8, 0$

models [31], CF2 and X2, fly as free rigid bodies: the attitude loop commands three body torques and a height loop the collective thrust. The upstream CF2 model exposes abstract moment channels [32]; we replace them with four rotor-force channels and the clipped inverse allocator (14), ordering rotors $(-x,-y), (-x,+y), (+x,+y), (+x,-y)$ with reaction signs $(-,+,-,+)$. Table I separates simulation parameters from manufacturer context; the CF2 $1.5\times$ scale raises a model limit, not measured hardware capability. Both models omit motor, RPM and battery states and validated rotor aerodynamics; feedback uses `qpos`/`qvel`, with noise injected only in Sec. IV-D.

CF2's body axes are principal axes. For X2 the geometry-derived body tensor, $\boldsymbol{J} = 10^{-2}[3.66517, 0, -.21000; 0, 2.54117, 0; -.21000, 0, 6.05280]$ kg m^2 (rows separated by semicolons), differs from both $\text{diag}(\boldsymbol{j}_{\text{tune}})$ and a manufacturer-identified tensor. Torque limits use $\boldsymbol{j}_{\text{tune}}$.

Each PD/CMG pair uses the same initial state, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic (no injected disturbances or sensor noise), with one run per cell and learning active throughout. We report the mean error $\bar{\theta}_q$ and relative reduction $100(\bar{\theta}_{q,\text{PD}} - \bar{\theta}_{q,\text{CMG}})/\bar{\theta}_{q,\text{PD}}$; means use all 90,000 samples even when plotted traces are decimated.

The five reference triples are generated as follows (radians): Roll is $(0.25 \sin 0.7t, 0, 0)$; Full3 is $0.5 \sin(t + (0, \pi/2, \pi)^\mathsf{T})$, componentwise; Hard45 is $(0.55 \sin 0.75t + 0.20 \sin(2.10t + 0.40),\ 0.45 \sin(0.95t + \pi/3) + 0.18 \sin 1.75t,$ $0.65 \sin(0.55t + \pi) + 0.25 \sin(1.35t + 0.20))$. Hardstep holds zero for one second, then cycles every two seconds through $(.75, -.45, .55)$, $(-.65, .55, -.70)$, $(.50, .50, .90)$ and $(-.80, -.35, .20)$. Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. Hover is omitted: its near-zero error admits no meaningful percentage.

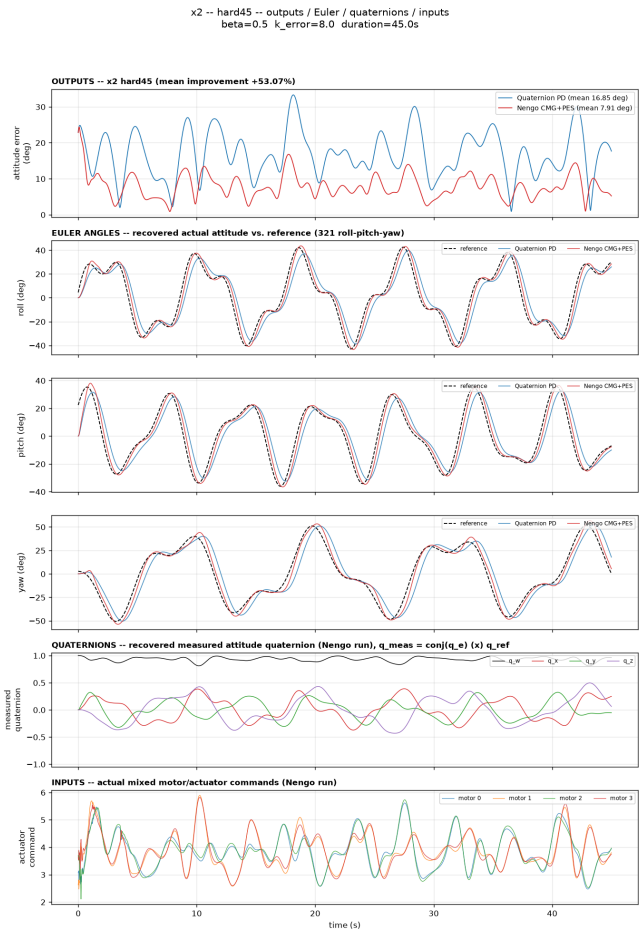


Fig. 4. X2 Hard45: PD baseline versus CMG–PES over the 45-s run. Hard45 drives all three axes at once and gives a large reduction that is also insensitive to measurement noise (Table II).

B. Baseline Comparisons

Table II collects every cross-mission result. Against the fixed PD law of (11), CMG–PES lowers mean quaternion-angle error in all ten nominal cases, by 17.3–60.6%, while sharing the actuator interface: rotor commands reach a bound only on Hardstep ($S = .016/.239$ on X2 and $.137/.264$ on CF2, PD/CMG). The largest reductions occur on Yaw360 and Full3 and the smallest on Roll. Holding the decoder at $W_k = W_0$ recovers at most a fifth of that gain, and the reductions survive both synthetic noise levels; Yaw360 is the one cell that loses appreciably, because its error approaches the noise floor.

Table II also reports four other neuron families driven by the same baseline and bounded interface; Sec. IV-E reads that comparison.

Figures 4 and 5 pair tracking performance and internal CMG signals for X2 Hard45. Improvement refers to the mean

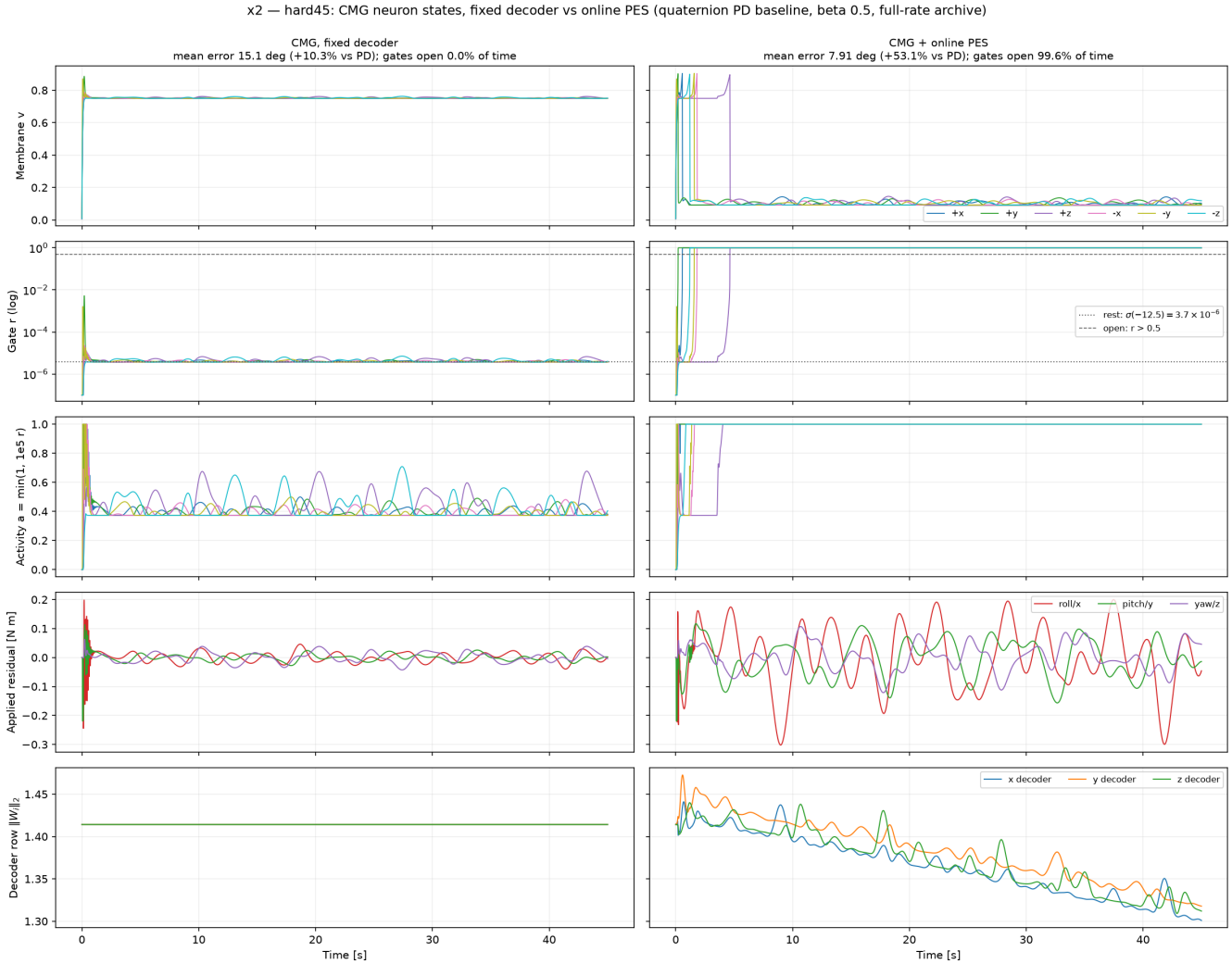


Fig. 5. X2 Hard45: internal CMG signals on the trajectory of Fig. 4, comparing fixed-decoder CMG (left) to online PES adaptation (right). Rows separate membrane state, raw gate r_j, and the clipped activity a_j presented to the decoder.

metric; it need not hold at every instant.

C. Stronger Baselines: Feedforward and Bandwidth

Equation (11) damps absolute rate, so on a moving reference it carries a velocity lag that a residual can remove without learning anything about the plant; repeating all ten cases on (12) separates the two effects. Feedforward alone removes 76–99% of the baseline error on the four moving-reference missions, and sweeping ρ over $[0.75, 8]$ minimises mission-mean error at $\rho = 3$ on both airframes and both noise levels, so the gains of (11) sit a factor of three below the optimum of their own law. Rotor authority sets that ceiling: CF2 diverges from $\rho = 4$ upward, a rotor at a bound in 83% of samples.

The residual improves every rung of that ladder, by a shrinking margin: $44.5/44.2\%$ over PD, $57.8/57.6\%$ over feedforward PD at $\rho = 1$, and $35.6/6.5\%$ at $\rho = 3$ (X2/CF2, geometric means). Two readings deserve emphasis. The best controller of the study is CMG–PES on the tuned baseline, on both airframes. Yet a tuned feedforward PD by itself already beats the PD-referenced augmented controller in every cell, by 13–300× on the smooth missions, so the 17.3–60.6% headline describes (11), not the best classical law of this family. Realized effort falls as the baseline improves: residual RMS on X2 Full3 drops from .121 to .0216 to .0148 N m across the three rungs.

Hardstep resists this accounting for a kinematic reason. It commands 63–111° rotations every 2 s, so 97–99% of its mean error is slew transient no bounded residual can remove, and the settled error is already 3.4° (X2) and 0.7° (CF2). Scored by time to enter a 5° band, the residual delivers what the mean hides: at $\rho = 1$ settling falls from 1.24 to 0.86 s on X2 (−31%) and 1.18 to 0.68 s on CF2 (−42%). Feedforward changes nothing there, since a piecewise-constant reference has $\boldsymbol{\omega}_d \equiv \mathbf{0}$.

At $\rho = 3$ the sign reverses, and Hardstep is the only cell in the study where the residual degrades performance. The mechanism follows from the latched decoder of (35): the applied residual is a first-order filter of its target with a 9.6 rad/s corner, so a loop running at $\rho\omega_n = 12$ rad/s outruns it by 72° of phase. Meanwhile the stiffer baseline saturates the torque box on 14–45% of samples (X2) and

69–94% (CF2). On a saturated axis the interaction is one-sided: a residual that agrees with $\boldsymbol{u}_0$ is clipped away and does nothing, while one that opposes it subtracts up to $\beta\ell_i$ of authority precisely when the loop needs all of it. The stale residual therefore brakes the slew, and settling rises from 0.59 to 0.83 s on X2.

That reading suggests a remedy: suppress the residual on an axis that is both saturated and opposed, $s_i \leftarrow 0$ when $|u_{0,i}| = \ell_i$ and $\operatorname{sgn} s_i \neq \operatorname{sgn} u_{0,i}$. It leaves $|\boldsymbol{u} - \boldsymbol{u}_0| \leq \beta\boldsymbol{\ell}$ and exact baseline recovery untouched, changes nothing while the baseline stays inside its box, restores settling to 0.57 s (X2) and 0.51 s (CF2), below the PD baselines, and keeps $\beta = 0.5$ admissible on CF2 where the ungated residual diverges. The test covers only the saturated $\rho = 3$ Hardstep cells; a systematic study under noise, disturbance and mismatch is future work, and all results reported here use the ungated law.

D. Sensitivity to Measurement Noise

Each of the ten cases was rerun at two synthetic noise levels with two paired seeds: at every step independent zero-mean Gaussian samples form a small Euler-angle quaternion left-multiplied into q, and additive samples corrupt each rate component. The moderate level is 0.010 rad and 0.025 rad s^{-1}, the realistic level 0.020 rad and 0.050 rad s^{-1}; neither is a calibrated sensor specification. The runner evaluates θ_q from the *corrupted* quaternion the controller uses, not from an independent ground truth, so the noise rows of Table II measure noisy-feedback error and inherit its floor.

All noisy reductions stay positive, spanning 18.08–55.75% at the higher level; Yaw360 loses about 12.8 percentage points against clean feedback, its error being closest to that floor, whereas Roll's reduction increases. Seed standard deviations are .001–.101 pp (moderate) and .002–.031 pp (realistic): limited repeatability, not a confidence interval.

E. Ablation: What Drives the Improvement?

We separate decoder adaptation (PES), input-dependent conductance modulation (χ), and readout gain (G). Toggling χ, which retains the base shunt $g_0 r_j^2$, is numerically indistinguishable with the decoder mode held fixed. *Fixed versus adaptive decoder.* Holding $W_k = W_0$ keeps the neuron dynamics but reduces the readout to $\boldsymbol{a}^+ - \boldsymbol{a}^-$, a bounded, stateful image of the PD request. It still helps (X2 0.12–10.27%, CF2 0.06–6.80%; Table II), so fixing the decoder does not remove *all* improvement, yet it recovers at most a fifth of the PES gain in any cell. Fig. 5 shows why, row by row. *Membrane and gate.* At $I_j = I_b$ the membrane rests at $v_j \approx 0.75 < v_{\mathrm{th}}$, so $r_j = \sigma(\alpha_h(I_b - v_{\mathrm{th}})) = \sigma(-12.5) \approx 3.7 \times 10^{-6}$: the gate is deep in its logistic tail, not closed. Once $r_j \geq 1/2$, $\gamma = 2$ and $v_j \geq 0$ (Proposition 1) give $b_{j,k} \geq 1/2$ in (21), so an opened gate cannot close (none did in the twenty runs), and the shunt holds v_j near 0.09. *Activity.* With $G = 10^5$ the resting gate already gives $a_j \approx 0.37$, graded by the input; a latched gate pins $a_j = 1$. With the fixed decoder only the $\pm y$ pair latches on Full3 (both models), so $a_y^+ - a_y^- = 0$ and the pitch residual is

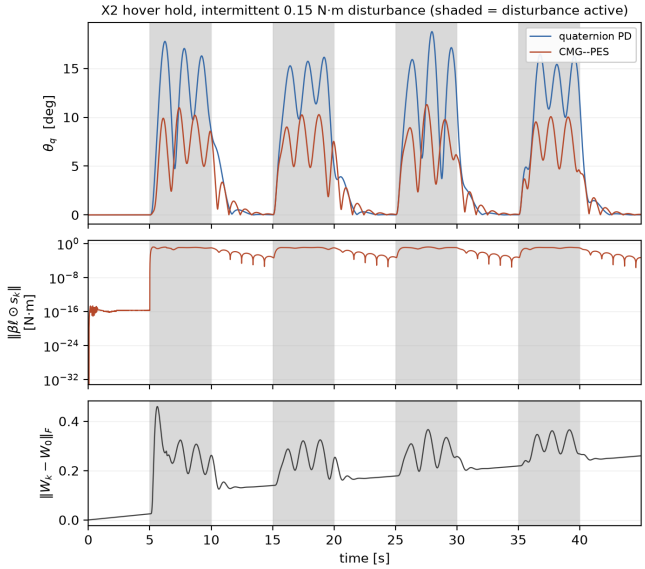


Fig. 6. X2 hover hold under four 0.15 N m disturbance bursts (shaded) separated by quiet intervals. Top: attitude error. Centre: residual magnitude, logarithmic. Bottom: decoder displacement from W_0. The residual is at numerical precision before the first burst. This means negligible initial correction, not zero computation or energy cost.

exactly zero after 1 s while the sub-threshold units carry roll and yaw. Under PES on X2, $\boldsymbol{a}_k \equiv \mathbf{1}$ from 2.02 s (all gates latched by 2.4 s), and for constant $\bar{\boldsymbol{a}}$ with inactive projection the direct update gives $\boldsymbol{z}_k = W_k\bar{\boldsymbol{a}}$ the recursion

$$\boldsymbol{z}_{k+1} = (1 - \lambda_W)\boldsymbol{z}_k + \eta\|\bar{\boldsymbol{a}}\|^2[\boldsymbol{y}_k^* - \mathrm{clip}_{[-1,1]}(\boldsymbol{z}_k)], \quad (35)$$

the leaky constant-feature regime of Proposition 3: a first-order filter of the teaching target with pole 0.9952 (about 0.104 s) and DC gain ≈ 0.9998, hence *lagged proportional* action rather than integral action, adding 1.50× (roll, pitch) and 1.63× (yaw) the PD stiffness behind a 9.60 rad/s corner. The references are sums of constant-frequency components [37] whose largest instantaneous frequency, 2.1 rad/s (Hard45), lies below that corner (gain 0.977, lag 12.3°). The same corner separates the bandwidths in Sec. IV-C. *Residual and decoder.* The residual RMS rises from 0.026 to 0.123 N m and the decoder moves $\|W_T - W_0\|_F = 0.37$; on Roll, Full3, Hard45, and Hardstep PES applies 4–12× the fixed residual RMS on both models. Where no gate opens (Roll, Yaw360; $r_j \leq 1.6 \times 10^{-3}$) PES lifts the driven membranes toward threshold ($v_j \leq 0.87$) and uses the graded tail. Decoder adaptation is therefore the principal lever: the CMG dynamics supply a bounded basis, latched or graded, on which PES shapes the correction. The leak λ_W is the σ-modification of robust adaptive control [38], [39], leaving an exact PES mismatch of $2.1 \times 10^{-4}\|\boldsymbol{s}\|$ latched and $1.5 \times 10^{-3}\|\boldsymbol{s}\|$ at the resting tail.

The Nengo benchmark spans two models, five missions and eleven controllers: PD plus fixed and PES variants of CMG, LIF, ALIF, Izhikevich [40] and LIFRate, all sharing the bounded residual, saturation and allocation interface, so only the neuron dynamics and the native-state-to-activity map differ. Fig. 7 shows two patterns. First, PES is the dominant lever in every family: fixed decoders rarely match

TABLE II

MEAN q-ERROR, 10^{-3} RAD, 45 S; Δ = REDUCTION OVER THE BLOCK'S OWN BASELINE [%]; EFFORT = RMS RESIDUAL AS A FRACTION OF THE PER-AXIS BOUND. CLOSING COLUMN: GEOMETRIC MEAN, ARITHMETIC FOR THE EFFORT ROWS.

	Skydio X2						Crazyflie 2, 1.5× rotor design					
	Roll	Full3	Hard45	Hardstep†	Yaw360	*geo*	Roll	Full3	Hard45	Hardstep†	Yaw360	*geo*
PD baseline, $\boldsymbol{\omega}_n = (4, 4, 3.5)$ *rad/s*												
PD	50.56	303.28	294.08	454.53	75.86	*173.12*	51.34	288.38	285.45	426.18	75.81	*168.68*
+CMG–PES	41.79	131.75	138.01	359.35	29.90	*96.02*	42.44	126.48	132.32	347.89	29.87	*94.11*
Δ [%]	17.4	56.6	53.1	20.9	60.6	*44.5*	17.3	56.1	53.6	18.4	60.6	*44.2*
Δ, fixed decoder [%]	1.9	9.7	10.3	0.5	0.1	*4.6*	3.3	4.5	6.8	0.4	0.1	*3.1*
Δ, moderate noise [%]	32.3	56.4	52.9	20.8	56.8	–	32.4	55.9	53.4	18.3	56.8	–
Δ, realistic noise [%]	32.2	55.7	52.3	20.6	47.8	–	32.5	55.2	52.7	18.1	47.8	–
Same baseline and bounded interface: strongest rival basis, and realized effort												
Δ, Izhikevich–PES [%]	51.1	52.3	49.7	22.6	60.7	*48.6*	51.5	51.9	50.3	21.4	56.4	*47.5*
Δ, Izhikevich fixed [%]	4.6	4.9	4.6	0.7	0.2	*3.0*	5.6	5.8	4.0	-2.2	-2.2	*2.3*
effort, CMG–PES	0.02	0.15	0.16	0.24	0.04	*0.12*	0.02	0.15	0.16	0.23	0.04	*0.12*
effort, Izhikevich–PES	0.38	0.35	0.36	0.39	0.38	*0.37*	0.37	0.36	0.36	0.39	0.37	*0.37*
PD + rate feedforward (12)*,* $\rho = 1$												
PD+ff	2.98	72.10	71.02	454.77	1.78	*26.19*	4.91	42.72	53.18	425.93	0.74	*20.39*
+CMG–PES	0.61	32.81	38.17	359.16	0.60	*11.04*	0.97	20.29	28.23	347.68	0.25	*8.65*
Δ [%]	79.5	54.5	46.3	21.0	66.4	*57.8*	80.2	52.5	46.9	18.4	66.4	*57.6*
PD + feedforward, $\rho = 3$ *(bandwidth-tuned)*												
PD+ff	0.34	10.09	10.54	265.91	0.20	*4.54*	0.56	7.06	8.65	262.59	0.084	*3.77*
+CMG–PES	0.074	8.83	9.08	311.07	0.12	*2.93*	0.43	6.96	8.50	265.93	0.079	*3.52*
Δ [%]	78.4	12.4	13.8	-17.0	41.8	*35.6*	22.9	1.3	1.8	-1.3	5.6	*6.5*

LIF, ALIF and LIFRate stay within 0.1% of the baseline outside Hardstep (best 18.5% on X2, 8.4% on CF2 there) at an effort of 0.03–0.04, so they are summarised here rather than tabulated. Noise rows are two-seed means, seed SD below 0.03 pp. †Hardstep commands 63–111° steps every 2 s, so 97–99% of its mean error is slew; Sec. IV-C scores it by settling time instead.

their PES counterparts, even with more expressive dynamics, extending Sec. IV-E to every basis. Second, lower error is not free: some Izhikevich–PES cells reach the smallest error, but at an average normalized residual RMS near 0.37 against 0.12 for CMG–PES, mission by mission in Table II, and with larger decoder movement, so equal output caps do not imply equal realized effort. Proposition 3 explains the flat rows: contraction scales with $1 - \eta\|\boldsymbol{a}\|_2^2$, so a basis that is silent stalls the update. LIF, ALIF and LIFRate units are active in 0.1% of unit-samples on Full3 and Hard45 and under 0.05% on Roll and Yaw360, whereas CMG (resting at $a_j \approx 0.37$ on its logistic tail) and Izhikevich are active in $\geq 99.9\%$. Persistent activity is therefore necessary for PES to act at all, but it does not by itself set effort, since Izhikevich shares it and still spends three times the authority. CMG–PES thus attains comparable tracking at markedly lower realized effort and more conservative decoder motion.

F. Disturbance Rejection and Retention

An equilibrium test applies $\boldsymbol{\tau}_\mathrm{d} = d_b \varrho(t)[\sin(1.31t + .2), .8\sin(1.73t + 1.1), .6\sin(2.17t + 2)]^\mathsf{T}$ in body axes, ϱ alternating 5-s on/off intervals after 5 s of quiet with 0.25-s raised-cosine ramps and $d_b = .15$ N m on X2 (the first component's amplitude). The four-burst 45-s run (Fig. 6) cuts burst-phase mean error from $11.50°$ to $6.65°$ (42.2%) and post-burst quiet error from $1.21°$ to $.99°$ (18.0%); the CF2 rerun ($d_b = .00015$ N m, a selected scaling) falls from .141199 to .070259 rad (50.24%).

V. DISCUSSION AND CONCLUSION

A bounded residual of six continuous CMG units and eighteen PES-adapted decoder weights augments a fixed quaternion PD baseline without changing its gains or actuator interface, reducing mean quaternion-angle error by 17.3–60.6% over ten matched 45-s MuJoCo cases on two modeled platforms, within rotor saturation and with exact baseline recovery at zero authority; direct and Nengo realizations agree to 2.013×10^{-6} relative mean error. Section IV-C places that headline: the residual improves a feedforward PD at the same gains by $57.8/57.6\%$ in geometric mean, every cell included, and a bandwidth-tuned one by $35.6/6.5\%$, yet that tuned law alone already beats the PD-referenced result and the residual costs $17.0/1.3\%$ on Hardstep there. It is complementary to a baseline missing a term, not a substitute for tuning one.

Decoder adaptation dominates in the tested regime: a fixed decoder gives at most 10.3%, and toggling χ is indistinguishable. Opened gates latch and saturate the readout, where a fixed paired decoder can cancel an axis, while sub-threshold units stay in a graded tail; PES shapes the correction on either basis. Across all five neuron families PES is the consistent differentiator, and the dynamics mainly set an effort–accuracy tradeoff. The readout gain G, which sets the resting activity at 0.37, is a key sensitivity parameter.

Limitations: results are in simulation (Sec. IV-D bounds noise sensitivity; model mismatch and hardware effects are not addressed), the retention horizon is limited, and bounded commands, states, and weights are implementation properties, not global stability proofs.

Future work includes integral-action and matched-stiffness baselines, a systematic study of the saturation-gated residual of Sec. IV-C under noise, disturbance and model mismatch, matched-effort comparisons against non-neural bounded bases, sweeps of G and χ, longer retention studies, held-out evaluations with frozen learned weights, full 6-DOF extension, and, most importantly, validation on phys-

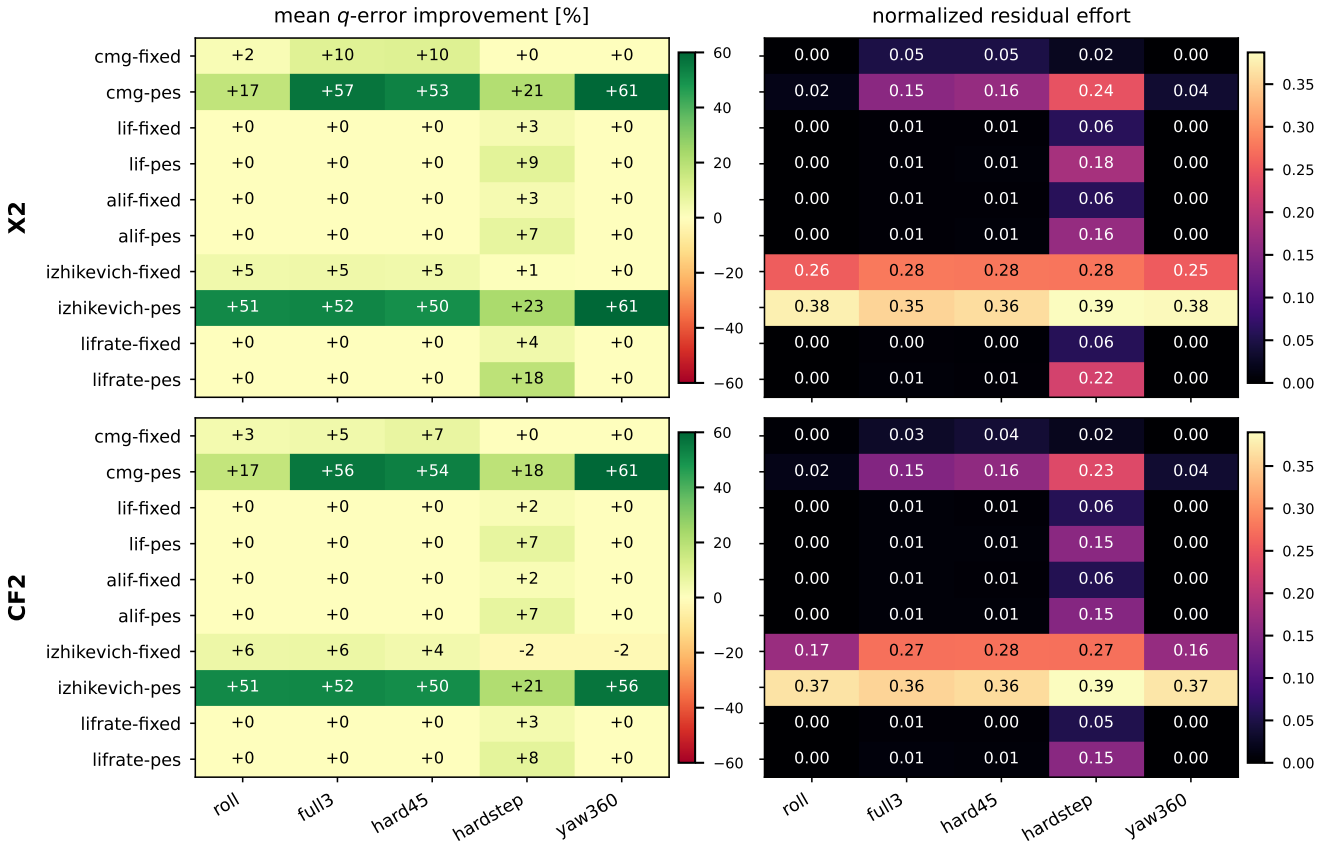


Fig. 7. Mechanism separation across the Nengo neuron-family benchmark, X2 (top) and CF2 (bottom). Left: mean q-error improvement over the PD baseline. Right: normalized residual effort, the RMS residual as a fraction of the per-axis limit. Rows are neuron family and learning mode, columns are missions; every cell uses the same bounded residual, saturation, and allocation interface. Two patterns hold on both platforms: PES adaptation, not the neuron model, separates the improving rows from the flat ones; and the families that match or exceed CMG–PES on error do so at three to twenty times its residual effort.

ical aerial, ground, and other robotic platforms; the same bounded-residual wrapper extends to nonlinear and optimal-control baselines with hard actuator constraints.

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